

# Design and Implementation of a 13.56 MHz Trace Antenna for an NFC Smart Business Card PCB

EECE 531: Electromagnetic Field Theory Final Project

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**Abstract**—This paper details the theoretical design, electromagnetic calculations, and physical layout of a high-efficiency 13.56 MHz passive Near Field Communication (NFC) business card. The project integrates a planar rectangular trace antenna onto a 0.8 mm FR4 printed circuit board (PCB) to facilitate wireless data transmission. By calculating the parasitic capacitance and matching the trace inductance to an NXP microchip, high-efficiency resonant coupling is achieved. Furthermore, aesthetic manufacturing techniques, including custom solder mask apertures and Electroless Nickel Immersion Gold (ENIG) surface finishes, are evaluated for their impact on structural integrity and RF performance.

**Index Terms**—Near Field Communication (NFC), Planar Trace Antenna, PCB Design, 13.56 MHz, Electromagnetic Coupling, Impedance Matching

## I. INTRODUCTION

### A. Motivation

As professional networking transitions from physical media to digital data exchange, Near Field Communication (NFC) provides a seamless bridge utilizing high-frequency (HF) electromagnetic induction. The objective of this project is to design a fully functional, passive NFC smart business card constrained to the standard physical dimensions of a credit card.

### B. Principles of Near Field Communication

NFC is a short-range radio technology enabling communication between devices held in close proximity ( $< 10$  cm). Operating at the 13.56 MHz Industrial, Scientific, and Medical (ISM) band, NFC systems are based on High Frequency (HF) Radio Frequency Identification (RFID) standards and support data rates up to 424 kbit/s.

The principal mechanism of NFC communication relies on inductive coupling between an initiator (reader) and a target (tag). The initiator generates an alternating magnetic field. When the tag enters this field, the loop antennas couple electromagnetically, effectively forming an air-core transformer. The reader's antenna serves as the primary winding, inducing current in the tag's antenna (secondary winding), as illustrated in Fig. 1.

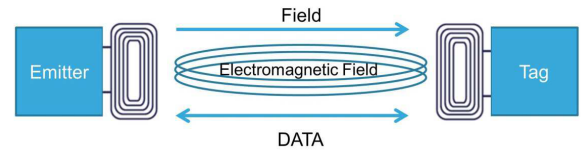


Fig. 1: Illustration of the inductive coupling mechanism between an active NFC reader and a passive NFC tag operating at 13.56 MHz.

### C. Environmental and Practical Advantages

A significant engineering advantage of passive NFC tags is their ability to operate without an internal power source. Because the integrated circuit harvests energy directly from the reader's magnetic field, the system is entirely battery-free. This eliminates the need for polluting button-cell batteries, resulting in a sustainable, maintenance-free electronic ecosystem.

## II. THEORETICAL BACKGROUND & ANTENNA DESIGN

### A. Tag Equivalent Circuit and Tuning

The design procedure originates from the base electrical model of the physical antenna coupled to the NFC integrated circuit, as depicted in Fig. 2.

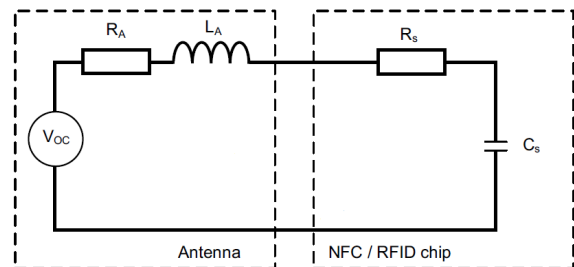


Fig. 2: Base electrical model of the passive NFC system, separating physical antenna properties from internal chip characteristics.

In this model,  $V_{oc}$  is the open-circuit voltage induced by the reader. The physical trace antenna is modeled as an ideal

inductor ( $L_{ant}$ ) with a parasitic loss resistance ( $R_A$ ). The NFC IC is represented by its series equivalent resistance ( $R_s$ ) and built-in tuning capacitance ( $C_s$ ). To simplify analysis, these are summed into the equivalent resonant LC tank circuit shown in Fig. 3.

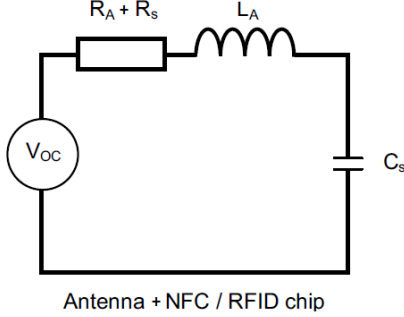


Fig. 3: Equivalent resonant circuit of the passive NFC tag.

The total equivalent capacitance ( $C_{eq}$ ) is the sum of any external tuning capacitor ( $C_{ext}$ ) and the IC's internal parallel capacitance ( $C_{IC}$ ):

$$C_{eq} = C_{ext} + C_{IC} \quad (1)$$

For the NXP NTAG NT3H1101 IC,  $C_{IC}$  is specified as 50 pF. Given the 13.56 MHz operating frequency ( $f$ ), the target trace inductance ( $L_{ant}$ ) is isolated via the LC resonance formula:

$$L_{ant} = \frac{1}{C_{eq}(2\pi f)^2} \quad (2)$$

Substituting 50 pF and 13.56 MHz into Equation (2) yields:

$$L_{ant} \approx 2.75 \mu\text{H} \quad (3)$$

### B. Planar Rectangular Inductance Geometry

The critical layout parameters governing  $L_{ant}$ —including outer dimensions ( $a_0$ ,  $b_0$ ), average dimensions ( $a_{avg}$ ,  $b_{avg}$ ), trace thickness ( $t$ ), width ( $w$ ), gap ( $g$ ), and number of turns ( $N_{ant}$ )—are defined in Fig. 4.

For this topology,  $L_{ant}$  is approximated using the modified Wheeler expression:

$$L_{ant} = \frac{\mu_0}{\pi} \cdot [x_1 + x_2 - x_3 + x_4] \cdot N_{ant}^{1.8} \quad (4)$$

where  $\mu_0 \approx 1.2566 \times 10^{-6}$  H/m, and variables  $x_1$  through  $x_4$  are derived from:

$$d = \frac{2 \cdot (t + w)}{\pi} \quad (5)$$

$$a_{avg} = a_0 - N_{ant} \cdot (g + w); \quad b_{avg} = b_0 - N_{ant} \cdot (g + w) \quad (6)$$

Applying optimized parameters ( $a_0 = 50$  mm,  $b_0 = 37$  mm,  $w = g = 0.3$  mm,  $N_{ant} = 5$ ) yields  $L_{ant} \approx 2.76 \mu\text{H}$ . This closely aligns with the  $2.75 \mu\text{H}$  target required by Equation (2).

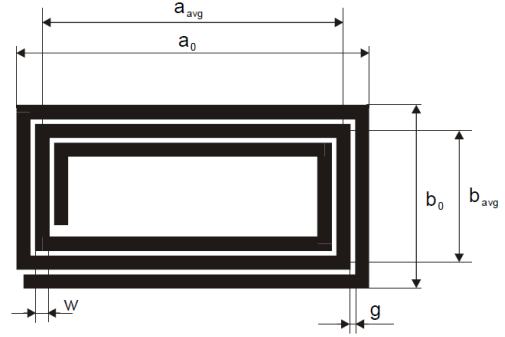


Fig. 4: Geometric layout parameters of the rectangular planar trace antenna.

### C. NXP Class 4 Reference Geometry Validation

To benchmark the design, an NXP Class 4 rectangular topology was evaluated. The Class 4 specification dictates an external boundary of 50 mm  $\times$  27 mm. NXP provides pre-simulated reference files for this geometry (Fig. 5), which served as a benchmark to validate the custom layout's  $L_{ant}$  performance.

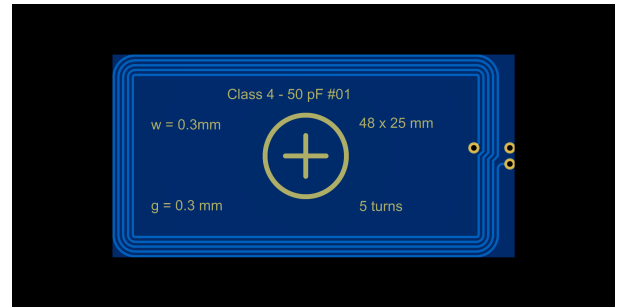


Fig. 5: The NXP Class 4 open-source reference antenna geometry.

## III. HARDWARE IMPLEMENTATION & PCB LAYOUT

### A. Integrated Circuit Selection (NXP NTAG I<sup>2</sup>C)

The NXP NTAG NT3H1101 was selected for its miniature SMD package and energy harvesting module. This IC bridges a passive NFC Forum-compliant contactless interface with a wired I<sup>2</sup>C contact interface. As shown in Fig. 7, the IC rectifies the reader's magnetic field to power internal logic and drive external peripherals.

### B. Schematic Synthesis and Component Integration

The schematic (Fig. 6) follows the pinout detailed in Fig. 8.  $L_{ant}$  is coupled to pins LA and LB. Energy harvested at VOUT (Pin 7) is routed to VCC (Pin 6), with a 220 nF capacitor ( $C_1$ ) for voltage stability.

The series current-limiting resistor ( $R_1$ ) for the LED was calculated via Ohm's law:

$$R_1 = \frac{V_{CC} - V_{LED}}{I_{LED}} = \frac{3.2 - 2.25}{20 \cdot 10^{-3}} \approx 47.5 \Omega \quad (7)$$

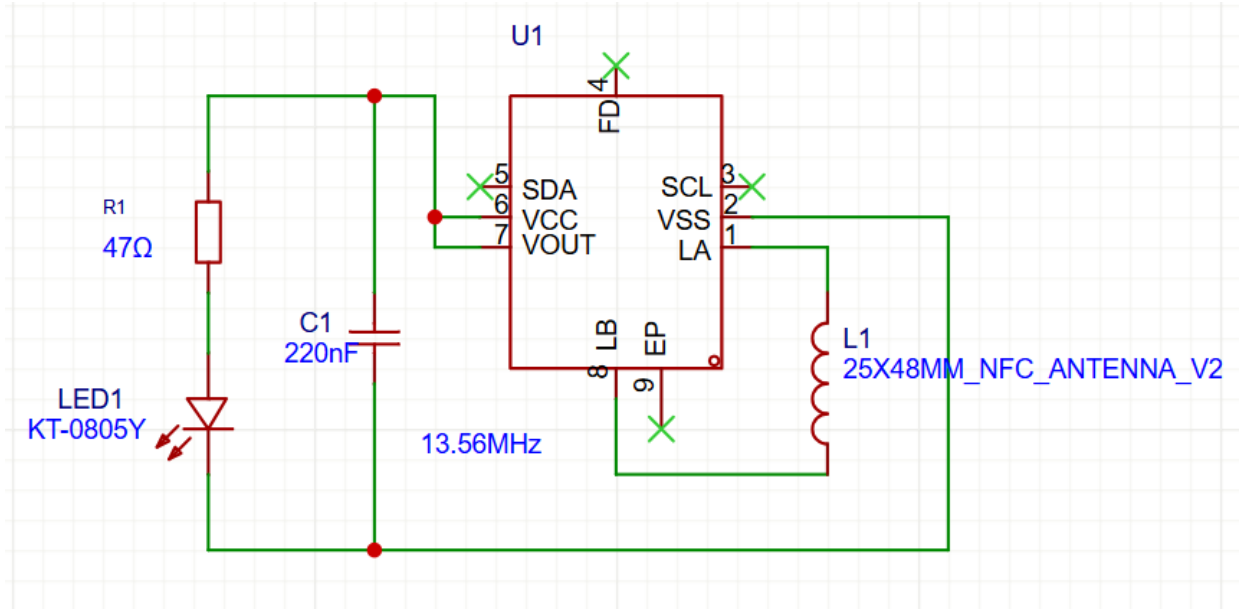


Fig. 6: Complete electrical schematic diagram of the NFC smart business card.

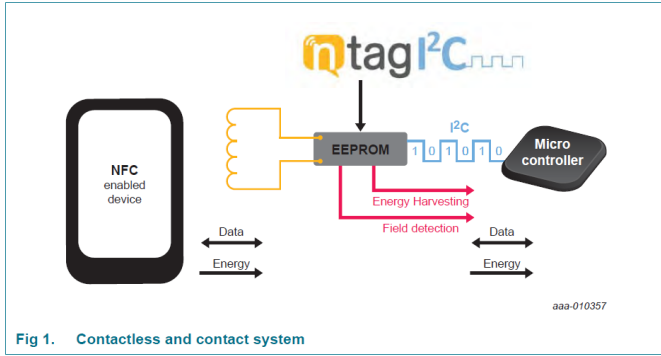


Fig. 7: Architecture block diagram of the NXP NTAG NT3H1101 IC.

| Pin | Symbol | Description                               |
|-----|--------|-------------------------------------------|
| 1   | LA     | Antenna connection LA                     |
| 2   | VSS    | GND                                       |
| 3   | SCL    | Serial Clock I <sup>2</sup> C             |
| 4   | FD     | Field detection                           |
| 5   | SDA    | Serial data I <sup>2</sup> C              |
| 6   | VCC    | VCC in connection (external power supply) |
| 7   | VOOUT  | Voltage out (energy harvesting)           |
| 8   | LB     | Antenna connection LB                     |

Fig. 8: Pinout description of the NXP NTAG NT3H1101 IC.

A 47  $\Omega$  resistor was selected. However, it is noted that harvested power is field-strength dependent; at greater distances,  $V_{CC}$  may sag under high current loads, potentially requiring a higher resistance to ensure IC stability.

#### C. Antenna Isolation and Ground Plane Strategy

Metallic planes act as shields that attenuate magnetic flux by inducing eddy currents. These currents create a counter-

magnetic field that opposes the reader's field, effectively "canceling" the coupling. To prevent this, strict copper keepouts were enforced. As shown in Fig. 9, maintaining a clear aperture adjacent to the inductive loop allows magnetic flux to pass through the substrate unimpeded.

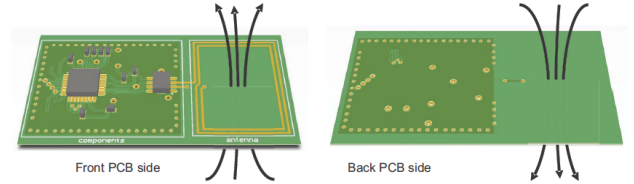


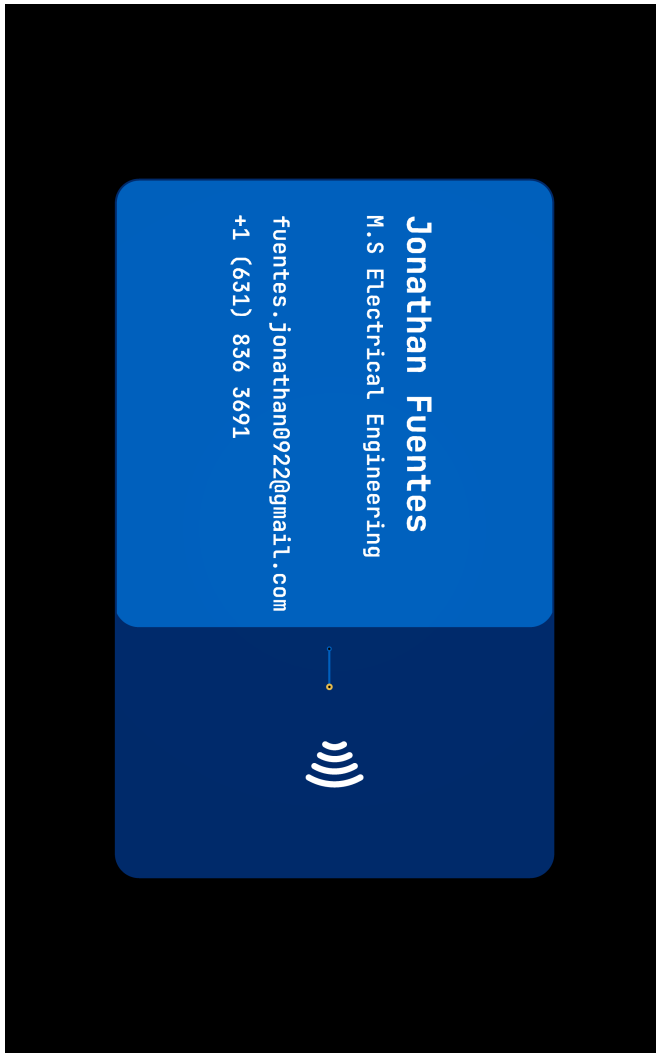
Fig. 9: Visualization of electromagnetic flux coupling and ground plane isolation.

#### D. Aesthetic Solder Mask Apertures

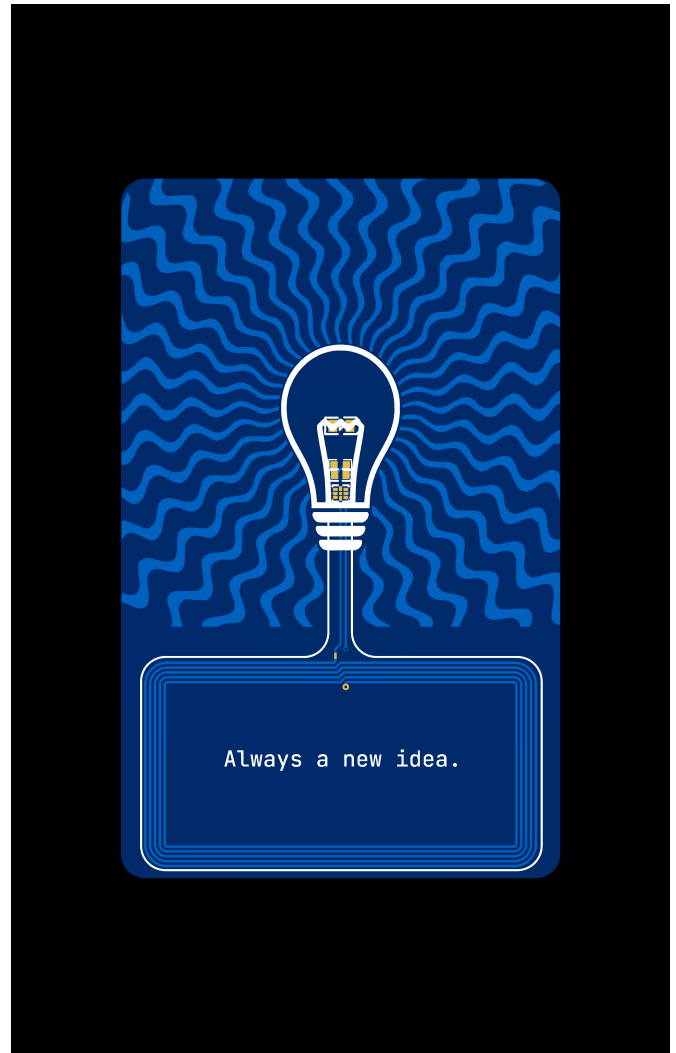
To achieve a shiny metallic gold effect for contact information, a non-traditional layering technique was employed. A partial ground plane was defined on the Top Layer, and text was imported onto the Top Solder Mask Aperture layer. This selectively withholds the mask, exposing the underlying gold-finished (ENIG) copper for a lustrous professional finish.

### IV. EXPECTED RESULTS & SIMULATION

The fabrication views of the finalized 0.8 mm FR4 board are presented in Fig. 10. The front view (Fig. 10a) illustrates the minimal footprint of the SMD components, while the back view (Fig. 10b) visualizes the aesthetic ENIG keepout structures.



(a) PCB Front Side (TOP) view.



(b) PCB Back Side (BOTTOM) view.

Fig. 10: Fabrication views of the passive NFC smart business card.

## V. CONCLUSION & FUTURE WORK

This project successfully maps electromagnetic field theory to a practical hardware design. Mathematical modeling of  $L_{ant}$  satisfies the conditions for 13.56 MHz resonant coupling. Future work will involve soldering the passive components and programming the NFC chip.

## REFERENCES

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